SenseEdge Rev. A Silicon Test Plan

Measurements for the CICC → JSSC paper

Fidel Makatia Omusilibwa

October 2026 (companion to `plan/plan.tex` and `paper/main.tex`)

Bottom line

What revision A can prove, to journal standard. (1) Acquisition, FFT, and feature extraction that match the RTL bit for bit across supply, frequency, temperature, and at least 20 dice. (2) Energy: per clock, per classification window, minimum-energy point, leakage, and a phase-resolved breakdown. (3) A same-die comparison of the hardware pipeline against the same pipeline in software on the Caravel RISC-V core. (4) Signal-chain fidelity against reference instrumentation. (5) Accuracy and a multi-week deployment on physical faults, with the features computed in hardware and the classifier in firmware on the same die.

What it cannot prove. The on-chip network cannot be programmed: 22 of its 212 weights are writable and the rest hold their power-up state. A search over the 22 writable weights does not rescue it. In simulation over 40 random power-up states, on a task that a fully loadable network solves at 100 %, the median accuracy after the search is 37 % and no state reaches 90 %. The claims "no processor in the inference loop" and "in-field model update through the register map" therefore cannot be measured on this silicon.

The operating point that matters. The clock is not gated, so energy per classification is set by the number of clocks per window, $2240\,(D+1)$ for divider $D \geq 1$ and 2304 at $D = 0$. At the reset divider and 20 MHz that is 0.36 mJ. With $D = 0$ and the clock set to 36 times the sample rate it is predicted at 1.5 µJ, with an average macro power of 54 µW at 2.29 kS/s. RTL simulation gives identical FFT bins and features at $D = 0$. The campaign is built around measuring this point.

Decision needed now (gate G0). Either the paper is reframed around what revision A measures, or the JSSC version waits for a corrected revision B. Section 1 lists every claim in the draft that changes.

1 Claims in the draft against what can be measured

Each row is a statement in `paper/main.tex`. "Tests" refers to Section 4.

Claim in the draft	Status on revision A	Tests	Action
Acquisition, FFT, features, classification, and alarm "entirely in hardware"	Acquisition, FFT, features: yes. Classification and alarm: the network runs but its weights are not programmable	B1–B5, N1	State where the classifier runs. Hardware classifier is a revision B result
"X µJ per classification at X MHz, X V"	Measurable. The network executes on every window, so its energy is included	C2, C4, C5	Define a classification as one 64-sample window; report D and f_{clk} with every number
"Duty-cycled average node power at one classification per second"	No on-chip gating. Duty cycling means stopping or slowing the external clock	C4, E3	Report clock policy. Node board needs a controllable clock source (Appendix A)
"$XX.X$% accuracy on four physical fault classes, X-week deployment"	Measurable with hardware features and the classifier in firmware	F1–F5	Report as hybrid; give the ladder of F3 so the cost of the front end is visible
"In-field model updates through a register-mapped weight interface"	Not on revision A	N1	Firmware-table update only; register-map update moves to revision B

Claim in the draft	Status on revision A	Tests	Action
"Samples a 12-bit ADC at up to 100 kS/s"	ADC serial-clock rating limits the rate; expect about 44 kS/s at 3.3 V	D1	Replace with the measured maximum
"One acquisition-to-alarm pass takes 11 924 cycles"; "latency 596 µs"	11 924 is one window at $D = 4$ in simulation. Buffer-full to decision is 729 clocks	B4	Replace: 729 clocks (36.5 µs at 20 MHz); window $2240\,(D + 1)$ clocks, 2304 at $D = 0$
"Closing timing at 20 MHz"	Register-to-register slack +7.0 ns at the slow corner; port paths violate on paper	C3	Report measured $f_{\max}$ and the port-path caveat
"Bit-exact FFT/NN cross-checks against Python references …gate-level simulation with SDF"	The earlier network reference was not bit-exact; gate-level runs are functional, not SDF	B2, B3	Cite the netlist-accurate reference and the post-route equivalence run
Energy decomposition "using per-block enables"	There is one enable. Blocks run in sequence	C6	Decompose by time-resolved current
"∼50× frequency and X× energy step" to 7 nm	Needs the measured 130 nm energy with the same definition of a classification	C5	Fill after C5; state both operating points

2 Pre-silicon predictions to be correlated

Every measurement is compared with a number fixed before silicon. The predictions come from RTL and post-route simulation and from the final hardening run (`proposal/data/`). A model-to-hardware table built from this list goes in the paper (test C9).

Quantity	Prediction	Test
Buffer full to decision	729 clocks: FFT 483, features 37, network 209	B4
Clocks per window	$2240\,(D + 1)$ for $D \geq 1$; 2304 at $D = 0$ (36 per sample)	B4
Sample rate	$f_s = f_{\text{clk}}/[35\,(D + 1)]$ for $D \geq 1$; $f_{\text{clk}}/36$ at $D = 0$	B4, D1
Capture	$2\,(c - 2048) + c_1$ within ±1024 codes; wraps outside	B1
Bands as computed	bins 2–5, 6–11, 12–21, 22–31 plus DC; peak one bin low	B3
Weight registers reachable	22 of 212	N1
Macro power, 20 MHz, 1.8 V, 25 °C	12.9 mW (7.2 mA on `vccd2`); 10.5 to 15.0 mW over corners	C2
Energy per clock, 1.8 V	0.65 nJ; 88 % clock network and flip-flop clock pins	C2, C6
Leakage	0.7 µW typical at 25 °C; 0.34 mW slow corner at 100 °C	C1
Energy per classification	1.5 µJ at $D = 0$; 0.36 mJ at reset; 0.47 µJ for the 729-clock pass alone	C4, C5
Longest register-to-register path	22.1 ns typical, 41.8 ns slow: $f_{\max} \approx$ 42 MHz typical, 23 MHz slow	C3
Hold margin, register to register	+0.06 ns at the fast corner	C3 (hold signature)

The leakage prediction matters for the slow-clock mode: at 100 °C it exceeds the 54 µW of that mode, so the energy per classification is expected to rise several-fold with temperature. C1 and C8 measure it.

3 Test infrastructure (build before silicon)

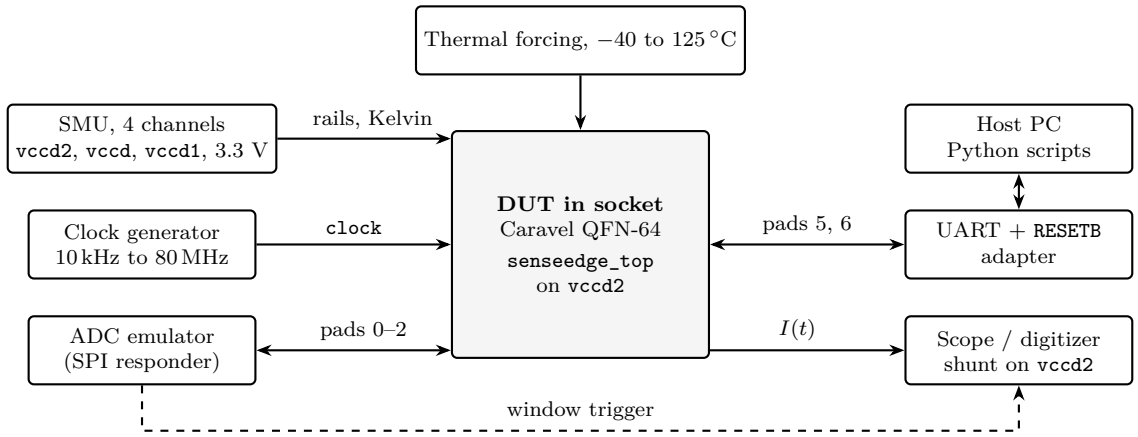


Figure 1: Characterization setup. The ADC emulator supplies known codes and marks window boundaries; the host drives every sweep and stores raw data.

I1. Characterization board

The sensor node ties the three 1.8 V pins together, has a fixed oscillator, and solders the ADC to the pads. None of the electrical tests can run on it. A separate board is required:

- QFN-64 test socket, so that all 50 parts can be measured.
- Separate Kelvin-sensed feeds for `vccd2` (macro), `vccd` (management core), `vccd1` (pad controls and the user-side enables of the isolation block; it must stay powered), and the 3.3 V rails. A shunt footprint in the `vccd2` feed with bulk decoupling on the supply side only.
- 50 Ω clock input to the `clock` pin. The harness passes the external clock straight to the core at reset.
- Header on pads 0–6, `RESETB`, the management GPIO, and the flash pins. Pads 1–4 are also the housekeeping SPI used to program the flash, so the header must give a programmer access to them.
- Footprint for the real ADC and an analog input, selectable against the emulator, for group D.

I2. Instruments

- Four-channel source-measure unit, current resolution 10 nA or better, digitizing at 100 kS/s or better on the `vccd2` channel.
- Clock generator, 10 kHz to 80 MHz, with a gated or burst mode and a 10 MHz reference output.
- Oscilloscope, 500 MHz, with a differential probe for the shunt.
- Thermal forcing for the socket, −40 to 125 °C, with a thermocouple at the package.
- Low-distortion signal source locked to the clock generator's reference.
- Reference vibration chain: IEPE accelerometer, 24-bit acquisition at 25.6 kS/s or more, tachometer, small shaker.

I3. ADC emulator

An FPGA or a microcontroller with a deterministic SPI responder that reproduces the MCP3201 frame (two sample clocks, null bit, twelve data bits, LSB-first repeat). Requirements: serial clock to 10 MHz; replay memory for at least 65 536 codes; state advanced only by the serial clock, so a paused chip resumes cleanly; a trigger output at the 64th sample of each window; codes loaded from the host.

I4. Firmware for the management core

- **Monitor.** Peek, poke, and burst read over the hardware UART (TX on pad 6, RX on pad 5). Pad 5 stays under management control; pads 0–4 are handed to the user area and the housekeeping SPI is disabled.
- **Self-check.** After each decision flag: clear enable, read the 32 bins and 8 features, compare a CRC with a table built from RTL simulation, count mismatches, set enable. No UART traffic per window.
- **Free run.** Chain enabled, nothing read; used for power.
- **Software pipeline.** The same capture arithmetic, FFT, features, and network in C, bit-exact to the netlist model, for test E1.
- **Firmware classifier.** The 8–16–4 network on features read from the chip, with weights in a table.

All register offsets are word offsets from `0x3000_0000`. The firmware in the repository does not address the chip correctly and is not a starting point.

I5. Golden data

The RTL is the golden model; the post-route netlist returns the same bins and features from three power-up states. Vector sets, generated by `proposal/src/run_sims.py`:

V1 Impulse records, one non-zero sample per window, stepping through all 4096 codes. Every bin then equals the magnitude of that sample, which reads the capture transfer out through the FFT.

V2 One tone per bin, 0 to 31, at three amplitudes.

V3 4096 random three-tone records with noise and offset.

V4 Corner cases: full-scale wrap, magnitude saturation, constant codes, alternating codes.

V5 Recorded vibration: the testbed recordings of group F and the CWRU set, resampled to the chip rate.

I6. Rehearsal

The firmware, emulator model, and host scripts run first against the full Caravel RTL in co-simulation and then against the user macro on an FPGA behind the same pads. No script meets silicon untested.

I7. Data discipline

Every record carries die number, board, firmware hash, script commit, instrument, calibration date, and ambient temperature. Raw data are kept; figures are generated by script from raw data. Dice are numbered on arrival and never re-numbered.

4 Tests

Predictions in *italics*. "Pass" in groups B and C means zero mismatches in self-check.

Group A: incoming and bring-up (all parts)

ID	Test	Procedure	Acceptance and record
A1	Continuity	Pin-to-rail resistance in the socket, unpowered	No shorts between rails
A2	Rails, quiescent	Ramp each rail with current limit, clock held low	Currents per rail; *microamperes on* `vccd2`
A3	Housekeeping	Read the housekeeping ID over SPI before any firmware runs	Life sign independent of firmware
A4	Boot	Program flash, release reset, UART banner	Boot at 10 MHz
A5	Pad handover	Configure pads 0–4 as user, 5–6 management; read pad states	Pads as configured

ID	Test	Procedure	Acceptance and record
A6	Register map	Read status; write and read back control, alarm, divider registers; time one read	Values match; read time sets the self-check divider
A7	First window	Emulator sends a bin-8 tone; read bins and features	Bit-exact to RTL. **Gate G1**

Group B: functional correctness, bit-exact (3 dice in full, all dice on V2)

ID	Test	Procedure	Acceptance and record
B1	Capture transfer	V1 at $D = 4$, 20 MHz; bin value against code	$\lvert 2(c - 2048) + c_1 \rvert$ *with wrap at* ± 1024; plot against the RTL curve
B2	FFT	V2, V3, V4; at least 10^5 windows	Zero mismatches against RTL
B3	Features	Same runs; feature read-back	Zero mismatches; band map as predicted
B4	Timing constants	Trigger at 64th sample; decision flag captured by a management timer; current signature as a second measure; $D = 0$, 1, 4, 249	*729 clocks; window* $2240\,(D + 1)$, 2304 *at* $D = 0$; identical results at every D
B5	Alarm and interrupt	Log class and confidence per window with the alarm pad and interrupt flag over 10^4 windows	Pad and flag follow the alarm model applied to the logged sequence
B6	Soak	Self-check on V3 for 10^8 windows at nominal, and 10^7 at $V_{\min} + 50$ mV	Zero errors; quote the 95 % upper bound $3/N$
N1	Network as fabricated	(a) Fixed records, 100 power cycles per die, 10 dice: log class and confidence. (b) Step each reachable weight, log the response. (c) Write the addresses above `0x74`; confirm no response	Documents the defect on silicon: repeatability of the power-up state within and across dice; *22 responsive registers*

Group C: electrical characterization

`vccd` and `vccd1` stay at 1.8 V unless stated; `vccd2` is the swept supply. Three dice receive every test at four temperatures. At least 20 dice receive the reduced set of C9.

ID	Test	Procedure	Acceptance and record
C1	Leakage	Clock stopped low and high; `vccd2` from 0.6 to 1.95 V; each temperature	$I_{\text{leak}}(V, T)$; 0.7 µW *at* 1.8 V, 25 °C
C2	Power against frequency	Free run; f_{clk} from 50 kHz to the limit, 12 points; 6 supplies; modes: chain disabled, chain enabled	Fit $P = E_{\text{clk}}(V) f + P_{\text{leak}}$; $E_{\text{clk}} = 0.65$ nJ *at* 1.8 V; enabled minus disabled gives the activity share
C3	Shmoo	Self-check, 256 windows per point; `vccd2` 1.95 to 0.8 V in 50 mV; f_{clk} 1 to 80 MHz; boundary points repeated with 4096 windows	Pass map; 42 MHz *at* 1.8 V, 25 °C. Failures that persist at low frequency and worsen cold and at high supply are hold failures: log separately
C4	Minimum-energy point	$D = 0$, $f_{\text{clk}} = 36 f_s$ for f_s = 1, 2.29, 4, 8 kS/s; lower `vccd2` to failure; energy per window from the digitized current over 1000 windows	$E_{\text{win}}(V)$ and $V_{\min}$ per rate; location of the minimum; 1.5 µJ *at* 1.8 V
C5	Energy per classification	Charge on `vccd2` over 1000 triggered windows at three points: reset configuration; $D = 0$ matched clock at nominal; the minimum of C4	Headline numbers with uncertainty; 0.36 mJ, 1.5 µJ

ID	Test	Procedure	Acceptance and record
C6	Phase-resolved energy	$f_{\mathrm{clk}} = 100\,\mathrm{kHz}$ so the pass lasts 7.3 ms; average the current over 1000 triggered passes; subtract the clock-only baseline	Energy of FFT, features, network, and acquisition; *split close to 483 : 37 : 209 because the clock dominates*
C7	Interface floor	Repeat the low-voltage edge of C3 with all three 1.8 V rails swept together	Separates core failure from level mismatch at the domain boundary
C8	Temperature	C1–C5 at −40, 25, 85, 125 °C	Energy per classification against temperature for both clock policies
C9	Die-to-die	At least 20 dice at 25 °C: leakage and power at 1.8 V; f_{max} at 1.6, 1.8, 1.95 V; V_{min} at $D = 0$, 2.29 kS/s; V2 bit-exact	Histograms, mean and standard deviation, functional yield of 50; model-to-hardware table against Section 2

Harness limits to state in the paper. The management core shares the clock, so the measurable f_{max} is capped by the core and its flash. If the macro still passes at that cap, report the cap and the $V_{\mathrm{min}}(f)$ curve below it. A Wishbone access to a macro below its working voltage never returns; the host must time out and pulse `RESETB`.

Group D: signal chain with the real ADC and sensor

ID	Test	Procedure	Acceptance and record
D1	ADC and capture	Bin-centred sine into the ADC, source locked to the chip clock, 8 amplitudes; raise the serial clock until bins degrade	Bin magnitude against input; spur level; noise floor per bin; maximum usable sample rate at 3.3 V
D2	Acceleration response	Node and reference accelerometer on the shaker; stepped sine 5 Hz to 3 kHz; three sample rates	LSB per g against frequency; noise floor in g; rejection of tones above half the sample rate
D3	Feature fidelity	Shaker multi-tone and motor signals; reference signal resampled and passed through the bit-exact model	Chip feature against reference feature: scatter, error histogram per feature
D4	Offset and orientation	Features with the sensing axis along and across gravity, before and after the offset remedy on the board	DC leakage into the high band, and its removal

Group E: system and baselines

ID	Test	Procedure	Acceptance and record
E1	Same-die software baseline	Software pipeline on the management core over V3; cycles from the core timer; energy from `vccd` minus its idle	Cycles and energy per classification, software against hardware, same die, same process, same vectors. Outputs must match bit for bit
E2	Microcontroller baseline	The same C pipeline with its own ADC on a commercial low-power microcontroller, measured with the same instrument	One row of the comparison table that is measured, not quoted

ID	Test	Procedure	Acceptance and record
E3	Node power	Sensor node, every rail, at 1, 2.29, 4 kS/s continuous and at 1 and 10 classifications per second with the clock stopped between windows. Bursts are exactly 2304 clocks at $D = 0$, aligned to the window, so each record is contiguous; the decision for a window appears in the next burst	Breakdown: sensor, ADC, macro, management core, oscillator, regulators, radio. One week on a stated cell, measured
E4	Hybrid decision cost	Firmware classifier on chip features: cycles and energy per decision	Cost of the classifier as it runs on revision A

Group F: classification and deployment on physical faults

ID	Test	Procedure	Acceptance and record
F1	Testbed	Motor with variable-speed drive and load; bearing housing with exchangeable bearings; imbalance disc with measured masses; adjustable base for measured misalignment. Reference accelerometer beside the node, tachometer	Fault severities stated in physical units; photographs; drawing
F2	Dataset	Healthy and three faults; two severities; three speeds; two loads; five reassemblies on different days; 2 min per session; chip features and reference signal recorded together	About 210 sessions. The session is the statistical unit, never the window
F3	Model ladder	On identical data: (i) chip features, INT8 8–16–4; (ii) chip features, floating point; (iii) designed bands recomputed from chip bins; (iv) 1024-point reference features with a stronger model; (v) RMS threshold. Leave-one-reassembly-out and leave-one-speed-out	Accuracy with 95 % interval from a session bootstrap; confusion matrix; what the 64-point front end, the shifted bands, and INT8 each cost
F4	Sample-rate policy	Fixed rate against a rate tied to shaft speed so that the shaft line sits in bin 2 or 3	Accuracy against policy; the shaft line is otherwise lost in bin 1
F5	Live run	Node classifies in real time for at least two weeks; faults introduced on a schedule unknown to the classifier; alarm depth swept offline on the log	Detection delay, false alarms per week against depth, drift against temperature, one firmware model update with before and after
F6	Second machine	A pump or fan; retrain; repeat F3 (i) and (iv)	Retargeting cost in data and accuracy
F7	Public benchmark on silicon	CWRU records replayed through the emulator; chip features; ladder (i), (ii)	A number others can reproduce, measured on the chip

5 What each test puts in the paper

Figure or table	From	Note
Chip micrograph with macro outline	—	Decapsulate one part or obtain a bare die; a layout plot does not replace it
Measurement setup, photo and diagram	I1–I3	
Capture transfer and spectrum, silicon against RTL	B1, B2	Bit-exact statement with window count

Figure or table	From	Note
Shmoo, three temperatures	C3, C8	Hold-type failures marked
Power against frequency; energy per clock against supply	C2	With the fit and the prediction
Energy per classification against supply, both clock policies	C4, C5	The headline figure
Phase-resolved current of one pass	C6	Replaces the "per-block enable" pie
Leakage and energy against temperature	C1, C8	
Die-to-die histograms	C9	$N \geq 20$
Model-to-hardware correlation table	C9	Predictions of Section 2
Chain response and noise floor	D1, D2	In physical units
Hardware against software on the same die	E1	Cycles and energy ratio
Node power breakdown; measured cell life	E3	
Confusion matrix; model ladder	F3	With intervals
Deployment timeline, false alarms against depth	F5	
Comparison table	C5, E2, F3	Energy with stated definition; one measured baseline row

6 Reporting rules

- Every number carries supply, clock, divider, temperature, and the number of dice. Energy per classification always states what a classification includes.
- Uncertainty is stated: instrument accuracy, shunt tolerance, and repeatability over five repeats. Die-to-die spread is reported separately from measurement uncertainty.
- Zero observed errors is reported as an upper bound, $3/N$ at 95 %.
- Accuracy is reported with an interval, on data split by session, with the split written down before the data are taken.
- The defects are reported: the weight decoder, the two signedness errors, the feature alignment, and the capture slice, with the silicon evidence of N1, B1, and B3.
- Scripts, raw data, vectors, and firmware are released with the paper.

7 Schedule and gates

When	Work	Gate
Now	Decide framing and revision B. Order socket, design characterization board. Revise the node board (Appendix A) before fabrication	**G0**: claims fixed
Before silicon	Board, emulator, firmware, host scripts; rehearsal in co-simulation and on FPGA; vector sets; testbed and reference chain built; F2 split written down	Rehearsal passes end to end
Weeks 1–2	A on all parts; B on three dice	**G1**: first window bit-exact. **G2**: B2, B3 clean
Weeks 2–4	C1–C7 on three dice at 25 °C; E1	Energy headline fixed
Weeks 4–6	C8 temperature; C9 twenty dice; B6 soak; N1	**G3**: silicon section complete for CICC
Weeks 3–8	D on the node; F1–F4 dataset and ladder; E2–E4	Accuracy with intervals

When	Work	Gate
Weeks 6–12	F5 live run; F6, F7	**G4**: deployment section complete for the journal version

8 Risks specific to this chip

- **Hold margin.** 0.06 ns at the fast corner. A die that fails at every frequency when cold or at high supply is a hold failure. Lower supply and higher temperature are the workaround; record which dice need it.
- **Port-path timing.** The reset and Wishbone input paths violate under the assumed arrival times. Symptom: wrong register writes or a chain that does not start cleanly after reset. A6 and A7 catch it.
- **Clock cap.** The management core may fail before the macro. Report it as a harness limit.
- **Domain boundary.** With `vccd2` below about 1 V the macro's outputs may no longer switch the 1.8 V receivers. C7 distinguishes this from core failure.
- **Flash access.** Programming needs pads 1–4, which the node wires to the ADC and LEDs. Without a header the flash cannot be reprogrammed in place.
- **Leakage at temperature.** It may dominate the slow-clock mode. If it does, report it; it is a result.

A Changes to the node board before it is fabricated

The node has not been assembled, so these cost a layout revision and not a rework.

1. Feed `vccd2`, `vccd1`, and `vccd` through separate jumpers with shunt footprints.
2. Make the clock controllable: oscillator through a jumper, a footprint for an external input, and the oscillator enable on a radio GPIO. A clock from the radio module at 36 times the sample rate, with the divider at zero, is the low-power mode.
3. Put a series jumper in the ADC data line and bring pads 0–4, `RESETB`, and the UART to a header, for the emulator and for flash programming.
4. Centre the sensor at mid-scale: AC coupling into a mid-supply bias, or an offset trim. Set the sensor bandwidth from the chosen sample rate; with the fitted 3.3 nF the bandwidth is 1.5 kHz, which aliases at every rate below 3 kS/s.
5. Swap the UART pins on the radio side: the harness transmits on pad 6 and receives on pad 5.
6. Check the 1.8 V regulator rating against C2. The macro alone is predicted at 7 mA at 20 MHz.
7. Populate the 20 MHz oscillator and one accelerometer part; the BOM file lists a crystal and a different sensor.

B Revision B, if the journal version is to claim the on-chip classifier

The same suite applies with three additions: weight read-back makes the network bit-exact test direct (B-group, all 212 registers); a register-map model update replaces the firmware-table update in F5; and clock gating adds a third clock policy to C4 and C5. Before a shuttle slot is committed the corrected RTL runs on the FPGA of I6 behind the node's front end, through groups B, D, and F.